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ORIGINAL ARTICLE

Proteomic Signatures of Cardiac Dysfunction Among People With Diabetes: The Atherosclerosis Risk in Communities Study

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BACKGROUND: To investigate the proteomic signatures of heart failure (HF) in diabetes. The underlying mechanisms of the elevated risk of HF in diabetes are unknown.

METHODS: In 10 189 ARIC study (Atherosclerosis Risk in Communities) participants free of HF (mean age 57±7 years, 56% women, 22% Black adults, 14% with diabetes), we conducted discovery and internal validation for the associations of 4955 plasma proteins with HF by diabetes status. We performed (1) Cox regression to identify proteins associated with HF by diabetes status, (2) external validation in the MESA study (Multi-Ethnic Study of Atherosclerosis, n=5233, 633 with diabetes), and (3) pathway analyses for identified proteins.

RESULTS: Over 24 years in ARIC, there were 2417 HF events (605 among individuals with diabetes). In 993 individuals with diabetes in the discovery sample, 19 proteins were associated with HF ($P<10^{-5}$), 12 proteins replicated in the internal validation sample ($P<0.05/19$). Six of the internally validated proteins replicated in MESA (false discovery rate, $q<0.05$). Five proteins were specifically associated with HF in those with diabetes: 4 are novel (inactive tyrosine-protein kinase 7, chondroadherin, leucine-rich repeat, and immunoglobulin-like domain-containing nogo receptor-interacting protein 1 and fibulin-5) and 1 is the previously known (cartilage intermediate layer protein 2). NPPB (N-terminal pro-BNP) was associated with HF in those with and without diabetes. Pathways over-represented among proteins associated with diabetes-related HF were lipid metabolism, inflammation, and brown adipose tissue (false discovery rate, $q<0.05$).

CONCLUSIONS: We identified 5 proteomic markers (4 novel) uniquely related to HF risk among individuals with diabetes and not among those without diabetes.

Key Words: cartilage ■ heart failure ■ leucine ■ proteomics ■ tyrosine

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Type 2 diabetes and heart failure (HF) are highly prevalent,[1,2] and strongly related.[3] Diabetes is a potent risk factor for HF, raising the risk by 2- to 3-fold.[3] HF has emerged as one of the leading cardiovascular complications of diabetes.[4,5] Preclinical studies in rodent models have helped describe some of the mechanisms (eg, glucotoxicity, lipotoxicity, oxidative stress, advanced tissue glycation, and hyperinsulinemia),[6,7] and molecular signatures (eg, O-GlcNAcylation, a major molecular cause of glucose toxicity) of adverse remodeling in the diabetic heart.[8] The

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WHAT IS NEW?

- There is limited proteomic data on the proteomic profile of heart failure related to diabetes. Therefore, we used a comprehensive, unbiased approach, profiling ≈5000 proteins, and identified the unique plasma proteomic profiles in individuals with diabetes-related patients with heart failure, as opposed to nondiabetes-related heart failure.
- The findings help elucidate the novel biology of diabetes-related heart failure, with the identification of 4 novel proteins that are highly specific to diabetes, including inactive tyrosine-protein kinase 7, chondroadherin, leucine-rich repeat, and immunoglobulin-like domain-containing nogo receptor-interacting protein 1 and fibulin-5.

WHAT ARE THE CLINICAL IMPLICATIONS?

- The identified proteins can be leveraged to improve heart failure risk prediction in the setting of diabetes, and thus help refine prevention strategies.
- The novel protein biomarkers of diabetes-related heart failure can help to identify modifiable biological pathways and potential druggable targets for heart failure.

Nonstandard Abbreviations and Acronyms

ApoM	apolipoprotein M
ARIC	Atherosclerosis Risk in Communities
BNP	brain natriuretic peptide
CHAD	chondroadherin
CHD	coronary heart disease
CILP2	cartilage intermediate layer protein 2
COL15A1	collagen alpha-1 (XV) chain
COLEC11	collectin-11
CST6	cystatin-M
CTSD	cathepsin D
FABP	fatty acid binding protein
FAP	prolyl endopeptidase
FBLN5	fibulin-5
GAL4	galectin-4
GDF	growth differentiation factor
GIF	gastric intrinsic factor
HF	heart failure
hs-cTnT	high sensitivity cardiac troponin T
LAG3	lymphocyte activation gene 3 protein
LINGO1	leucine-rich repeat and immunoglobulin-like domain-containing Nogo receptor-interacting protein 1
LXR/RXR	liver X receptor-retinoid X receptor
MESA	Multi-Ethnic Study of Atherosclerosis
MSMP	prostate-associated microseminoprotein
MYOC	myocilin 2
NPPB	N-terminal pro-BNP
NQO2	ribosyldihydronicotinamide dehydrogenase
NT-proBNP	N-terminal pro-brain natriuretic peptide
PTK7	inactive tyrosine-protein kinase 7
SCF	stem cell factor
SEMA4D	semaphorin-4D
SEMA6A	semaphorin-6A
SEMA6B	semaphorin-6B
TNF	tumor necrosis factor
TNFRSF11B	tumor necrosis factor receptor superfamily member 11B
TXNDC5	thioredoxin domain-containing protein 5

pathways linking diabetes to the development of HF in humans remain poorly characterized. Progress in refining our understanding of diabetes-related HF in humans has been hampered by the lack of specific biological signatures. Deep profiling using plasma proteomics can help identify novel proteins that reflect important pathophysiological processes and potential targets for therapies for diabetes-related HF.[6,7] However, few population-based studies have investigated the proteomic markers of diabetes-related cardiac dysfunction.[9–13]

Using data from the ARIC study (Atherosclerosis Risk in Communities), a population-based cohort of over 10 000 individuals with long-term follow-up, we investigated the associations of ≈5000 aptamer measurements of the plasma proteome with incident HF in people with and without diabetes. We used a split validation approach, including an exploration in a discovery sample (2/3 of the cohort) and a confirmation in a validation sample (1/3 of the cohort). We conducted an external replication of the top proteins in the MESA (Multi-Ethnic Study of Atherosclerosis). Additionally, we performed a pathway analysis to identify biological pathways and upstream regulators relevant to proteins associated with diabetes-related HF.

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METHODS

The data that support the findings of this study are available from the corresponding author on reasonable request.

Study Population

The ARIC study originally recruited 15 792 participants from 4 US communities—Minneapolis, MN; Jackson, MS; Forsyth County, NC; and Washington County, MD.[14] After the first visit in 1987 to 1989, participants have returned for subsequent visits and have also received telephone calls annually and semi-annually since 2012.

Our investigation focused on participants who attended visit 2 in 1990 to 1992 (aged 46 years to 70 years at the time).

Among the 14 348 participants who attended this visit, we excluded those with missing proteomic data (n=2537), nonfasting blood draw (n=382), prevalent HF (n=657), or missing covariates (n=492). Due to small numbers, we also excluded 42 participants who were neither white nor Black and 49 Black participants at the Maryland and Minnesota centers. Our final study sample of 10 189 individuals was randomly divided using diabetes as a stratifying variable into a 2/3 discovery sample (n=6854 [993 with diabetes]) and a 1/3 validation (n=3335 [484 with diabetes]) sample. This allowed us to conduct a discovery of proteins associated with HF among individuals with diabetes and then an internal replication of the identified proteins.

All study protocols received institutional review board approval at each study site, and all participants provided written informed consent.

Ascertainment of Diabetes Status

Diabetes status was ascertained at visit 2 based on a self-reported physician diagnosis, use of diabetes medications, a fasting blood glucose ≥126 mg/dL, or hemoglobin A1c ≥6.5%. Hemoglobin A1c was measured using high-performance liquid chromatography (Tosoh 2.2 Plus and Tosoh G7 analyzers, Tosoh Corporation, Tokyo, Japan).

Proteomic Profiling

Proteomic profiling was conducted in EDTA-plasma samples collected during ARIC Visit 2 (1990–1992) and stored at −80°C. In 2021, the relative abundance of 5284 plasma proteins was quantified using a highly multiplexed aptamer-based assay (SOMAScan Version 4, SomaLogic Inc, Boulder, CO) and expressed in relative fluorescence intensity units. The relative concentration of plasma proteins or protein complexes was measured using a Slow Off-rate Modified Aptamer (SOMAmer)-based capture array.[15] The protein abundances were standardized by SomaLogic using adaptive normalization by maximum likelihood techniques. The SOMAscan assay and its performance characteristics have been described in detail previously.[16,17] After the quality control exclusions, 4955 SOMAmers (4712 unique proteins) were investigated in our analyses.[18]

Covariates

Baseline (ARIC visit 2, 1990–1992) covariates included age, sex, race-center, current smoking, body mass index, total cholesterol, high-density lipoprotein cholesterol, systolic blood pressure, hypertension medication use, estimated glomerular filtration rate, NT-proBNP (N-terminal pro-brain natriuretic peptide), and hs-cTnT (high-sensitivity cardiac troponin T). Total and high-density lipoprotein cholesterol were assessed using enzymatic assays. Low-density lipoprotein cholesterol was estimated using the Friedewald equation.[19] Estimated glomerular filtration rate was calculated from serum creatinine and cystatin C using the 2021 Chronic Kidney Disease Epidemiology Collaboration equation.[20] Coronary heart disease (CHD) events were adjudicated. NT-proBNP was measured using an electrochemiluminescent immunoassay on an automated Cobas e411 analyzer (Roche Diagnostics, Mannheim, Germany). hs-cTnT concentrations were measured with a highly sensitive assay Elecsys troponin T (Roche Diagnostics, Indianapolis, IN) on an automated Cobas e411 analyzer.

Incident HF Assessment

Incident HF was defined as the first hospitalization or death related to HF occurring after Visit 2 (1990–1992), with follow-up through December 31, 2019. Hospitalizations and deaths due to incident HF were ascertained using the *International Classification of Disease*, *Ninth Revision* (*ICD*)-9 code 428 and *International Classification of Disease*, *Tenth Revision* code 150 from hospitalization or death certificates.[21]

Statistical Analyses

To identify proteins associated with diabetes-related HF, we stratified our analyses (in the discovery and validation samples) by baseline diabetes status. We $\log_2$-transformed protein abundance due to the right-skew and then standardized to facilitate comparisons across proteins.

We used Cox regression models to examine the association of each protein with incident HF. Person-time accrued from visit 2 (baseline, 1990–1992) to the date of HF hospitalization, date of death, or end of follow-up, whichever came first. Model 1 was unadjusted. Model 2 included age, sex, and race-center. Model 3 included all variables in model 2 plus the estimated glomerular filtration rate.[20] Model 4 included all variables in model 3 plus current smoking status, body mass index, systolic blood pressure, use of blood pressure medications, total-cholesterol, high-density lipoprotein-cholesterol, and CHD. Model 5 included all model 4 variables plus baseline log-NT-pro-BNP and log-hs-cTnT. The main results are from model 5.

In the discovery sample, we used a Bonferroni-corrected 2-sided $P<0.05/4955=1.009\times10^{-5}$ as the significance threshold. In the internal validation sample, the Bonferroni-corrected 2-sided P value used was based on the number of proteins identified in the discovery sample ($0.05/19=2.632\times10^{-3}$ among those with diabetes).

We tested for an interaction between diabetes status and each of the 12 internally validated proteins associated with HF among individuals with diabetes, by examining the P value from the likelihood ratio test of models with and without the protein×diabetes cross-product term and using a Bonferroni-corrected threshold of $<0.05/12=4.167\times10^{-3}$ to define statistical significance, recognizing tests for interaction have limited power. We additionally checked if the internally validated proteins associated with HF among individuals with diabetes were also associated with prevalent diabetes.

We used the R version 4.1.2 software for all analyses of ARIC data.

External Validation

We externally replicated our proteomic findings in the MESA.[22] This is a multiethnic, community-based prospective cohort study designed that enrolled 6814 men and women 45 to 84 years of age with no history of clinical cardiovascular diseases between July 2000 and August 2002. The MESA study participants (38% White, 12% Chinese-American, 28% Black, and 22% Hispanic adults) were recruited from 6 US sites: Baltimore, MD; Chicago, IL; Forsyth County, NC; Los Angeles, CA; New York, NY; and St Paul, MN.

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In the external replication analysis, we evaluated the proteins found to be associated with incident HF among 5233 MESA participants who attended the first MESA visit (2000–2002) and were free of HF. Of these participants, 633 had diabetes, with 85 developing HF. The latter was defined as an FPG ≥126 mg/dL, or self-reported diabetes or confirmed use of insulin or oral hypoglycemic medications at the index clinical examination, or a self-report of physician-diagnosed diabetes.

In MESA, using SOMAScan v4.1, over 7000 plasma proteins were quantified (inclusive of the 5000 proteins in SOMAScan v4.0). In MESA, the HF events were formally adjudicated. Two physician reviewers independently reviewed all available medical records for event classification and incidence dates and adjudicated the differences. If disagreement was not resolved after adjudication, the events committee makes the final classification of the event. Each potential HF event was categorized using standardized criteria as definite, probable, or no HF. A definite or probable HF diagnosis required thorough documentation of symptoms and signs. The MESA criteria for HF include (1) physician diagnosis of HF and patient receiving medical treatment for HF, (2) pulmonary edema/congestion on chest x-ray, (3) dilated ventricle or poor left ventricular function on echocardiography, radionuclide ventriculogram/multigated acquisitionor other contrast ventriculography, or evidence of left ventricular diastolic dysfunction.

The Cox regression models to examine the association of each protein and incident HF were adjusted for the variables included in the aforementioned model 5.

For external validation, the statistical significance was based on a false discovery rate (FDR) of $q<0.05$ in analyses stratified by diabetes status. We used the less stringent FDR approach as the MESA external validation sample was smaller than the ARIC discovery sample. We were interested in validating previously identified associations rather than in avoiding finding new false positive associations.

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Ingenuity Pathway Analysis for Proteins Associated With Diabetes-Related HF

We investigated the biological pathways underlying diabetes-related HF using the Ingenuity Pathway Analysis bioinformatic platform (QIAGEN Inc). This platform leverages the known associations between genes (or gene products) and regulators.[23] Among the 4995 proteins we evaluated, 4910 proteins were mapped to genes in the Ingenuity Pathways Knowledge base, which allowed us to characterize canonical (ie, biological) pathways and upstream regulators. As inputs, we used the set of proteins associated with incident diabetes-related HF at an FDR-corrected threshold of $q<0.05$. Analyses were conducted using the (Ingenuity Knowledge Base/full set of mapped SomaScan proteins as the reference set). IPA analyses included direct and indirect experimentally confirmed relationships derived from all species. We presented the top canonical pathways and upstream regulators that met the Benjamini-Hochberg FDR-corrected significance level.[23] The canonical pathways analysis identifies the biological processes and molecular functions for which there is an overrepresentation of proteins associated with diabetes-related HF. The upstream regulator analysis identifies molecules upstream of the proteins associated with incident diabetes-related HF.

RESULTS

Characteristics of the Study Participants

Among the 10 189 eligible participants (mean age 57±5.7 years, 56% women, 22% Black adults), 1477 (14.5%) had prevalent diabetes. Over a median follow-up of 24 years, there were 2417 HF events, including 605 among individuals with diabetes at baseline. Participant characteristics were similar in the 2/3 discovery and 1/3 validation samples (Table 1).

Discovery of the Proteomics Signature of Heart Failure

In the ARIC study 2/3 discovery sample (Table 1), there were 6854 participants, including 19% with diabetes (n=993). A total of 1618 incident HF events (432 among those with diabetes) occurred during a median of 24 years of follow-up.

Among participants with diabetes, there were 146 proteins associated (Bonferroni $P<0.05/4955$) with incident HF in unadjusted analyses (model 1, Table S1). This number was reduced to 125 proteins in model 2 (Table S2), 59 proteins after additional adjustment for estimated glomerular filtration rate (model 3, Table S3), and 41 proteins after additionally accounting for current smoking status, body mass index, systolic blood pressure, use of blood pressure medications, total-cholesterol, high-density lipoprotein-cholesterol, and CHD (Table S4). After adjustment for cardiometabolic risk factors, including CHD, NT-proBNP, and hs-cTnT (model 5), 19 proteins were associated with incident HF (Figure 1A; Table 2). The proteins that were negatively associated with incident HF were CILP2 (cartilage intermediate layer protein), COL15A1 (collagen alpha-1 [XV] chain), MYOC (myocilin 2), CST6 (cystatin-M), CHAD (chondroadherin), MSMP (prostate-associated microseminoprotein), NQO2 (ribosyldihydronicotinamide dehydrogenase; quinone), and FAP (prolyl endopeptidase). Positively associated proteins were NPPB (N-terminal pro-BNP), PTK7 (inactive tyrosine-protein kinase 7), SEMA4D (semaphorin-4D), LAG3 (lymphocyte activation gene 3 protein), SEMA6A (semaphorin-6A), SEMA6B (semaphorin-6B), COLEC11 (collectin-11), LINGO1 (leucine-rich repeat and immunoglobulin-like domain-containing nogo receptor-interacting protein 1), TNFRSF11B (tumor necrosis factor receptor superfamily member 11B), TXNDC5 (thioredoxin domain-containing protein 5), and FBLN5 (fibulin-5).

Among individuals without diabetes, there were 80 proteins were associated with incident HF in Model 5 (Table S5). Of these 80 proteins, there were only 4 proteins (6%) that were associated with HF (Bonferroni $P<0.05/4955$) in individuals both with and without diabetes (NPPB, CILP2, MYOC, and COL15A1).

Table 1. Baseline Characteristics of ARIC Study Participants at Visit 2 (1990–1992) According to Diabetes Status, Discovery and Replication Samples

	Discovery sample (N=6854)			Replication sample (N=3335)		
	Baseline diabetes	No diabetes	*P* value	Baseline diabetes	No diabetes	*P* value
N	993	5861		484	2851	
Age, y	58.2 (5.71)	56.8 (5.69)	*P*<0.01	57.8 (5.83)	56.7 (5.69)	*P*<0.01
Sex			0.24			0.10
Female	537 (54.1%)	3290 (56.1%)		250 (51.7%)	1591 (55.8%)	
Male	456 (45.9%)	2571 (43.9%)		234 (48.4%)	1260 (44.2%)	
Race			0.24			*P*<0.01
Black	392 (39.5%)	1120 (19.1%)		207 (42.8%)	525 (18.4%)	
White	601 (60.5%)	4741 (80.9%)		277 (57.2%)	2326 (81.6%)	
eGFR, mL/min per 1.73 m^2	96.35 (19.16)	98.78 (15.45)	*P*<0.01	96.16 (21.62)	99.02 (15.23)	0.005
Body mass index, kg/m^2	30.86 (5.71)	27.2 (4.92)	*P*<0.01	31.53 (6.06)	27.23 (4.88)	*P*<0.01
Current smoking	192 (19.3%)	1296 (22.1%)	0.05	85 (17.6%)	643 (22.6%)	0.01
Total cholesterol, mg/dL	214.23 (46.02)	209.2 (37.9)	0.02	213.84 (42.54)	208.43 (38.67)	0.007
HDL-cholesterol, mg/dL	43.70 (14.31)	51.04 (17.02)	*P*<0.01	43.70 (14.31)	51.04 (16.63)	*P*<0.01
Triglycerides, mg/dL	147.8 (105.3–209.7)	108.9 (79.7–153.1)	*P*<0.01	140.7 (101.8–198.2)	108.0 (78.8–153.1)	*P*<0.01
Systolic blood pressure, mm Hg	127.02 (18.47)	119.88 (18.23)	*P*<0.01	129.14 (21.82)	119.84 (17.89)	*P*<0.01
Use of blood pressure medications	443 (44.6%)	1267 (21.6%)	*P*<0.01	241 (49.8%)	568 (19.9%)	*P*<0.01
Diabetes	993 (100%)	0 (0%)		484 (100%)	0 (0%)	
HbA1c, %	7.67 (2.09)	5.4 (0.36)	*P*<0.01	7.75 (2.19)	5.4 (0.37)	
Fasting glucose, mg/dL	182.77 (80.97)	101.18 (9.41)	*P*<0.01	183.05 (78.65)	101.36 (9.32)	
Use of glucose-lowering medications	397 (40.1%)	…		205 (42.4%)	…	
NT-proBNP, pg/mL	44.82 (23.46–85.29)	53.17 (28.54–97.59)	*P*<0.01	44.63 (20.58–91.88)	52.69 (28.36–91.6)	0.076
hs-cTnT, ng/L	5 (1.5–8)	1.5 (1.5–5)	*P*<0.01	5 (1.5–9.25)	1.5 (1.5–6)	*P*<0.01
Prevalence of coronary heart disease	87 (8.8%)	272 (4.7%)	*P*<0.01	43 (9%)	122 (4.3%)	*P*<0.01

Data are mean (SD) or median (interquartile range) or n (%). CHD indicates coronary heart disease; eGFR, estimated glomerular filtration rate; HbA_{1C}, glycosylated hemoglobin; HDL, high-density lipoprotein; hs-cTnT, high-sensitivity cardiac troponin I; and NT-proBNP, N-terminal pro-brain natriuretic peptide.

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Internal Validation of Proteins Associated With Diabetes-Related Heart Failure

In the internal 1/3 ARIC study validation sample, there were 799 incident HF events (225 events among those with diabetes) during a median follow-up period of 24 years.

Of the 19 proteins associated with HF among individuals with diabetes (model 5), 12 proteins were also significantly (*P*<0.05/19) associated with incident HF among individuals with diabetes in the validation sample. For all these 12 proteins (CILP2, NPPB, COL15A1, PTK7, CHAD, SEMA4D, SEMA6B, SEMA6A, LINGO1, TNFRSF11B, TXNDC5, and FBLN5), the direction and magnitude of the effect estimates were concordant across the discovery and internal validation samples (Figures 1B and 2; Table S6).

Among the 12 internally validated proteins, in the internal validation sample, 5 showed a statistically significant interaction by diabetes status for the HF outcome (Tables S7). The proteins that had a significant *P* value for interaction (<0.05/12) were: COL15A1 (*P* interaction=1×10^{-5}), PTK7 (*P* interaction=2×10^{-5}), SEMA-4D (*P* interaction=3×10^{-5}), SEMA-6A (*P* interaction=1×10^{-5}), and LINGO1 (*P* interaction=3×10^{-5}). COL15A1 showed a stronger protective effect among those with diabetes, whereas PTK7, SEMA-4D, SEMA-6A, and LING01 were more strongly associated with HF among those with diabetes compared with persons without diabetes (Table 2).

Sensitivity analyses indicated that none of the 12 internally validated proteins (associated with HF among individuals with diabetes) showed an interaction with sex or race for HF risk (*P*>0.05). The associations for the 12 internally validated proteins were not statistically significantly different for events that occurred in the first decade of follow-up or subsequent decades of follow-up (*P*>0.05).

Of the 12 internally validated proteins associated with diabetes-related HF, 9 of these proteins were also associated with prevalent diabetes (Figure 2; Table 3): CILP2, COL15A1, PTK7, CHAD, SEMA4D, SEMA6B, SEMA6A, LAG3, and TXNDC5. All of these associations had the same direction for diabetes as the associations with HF.

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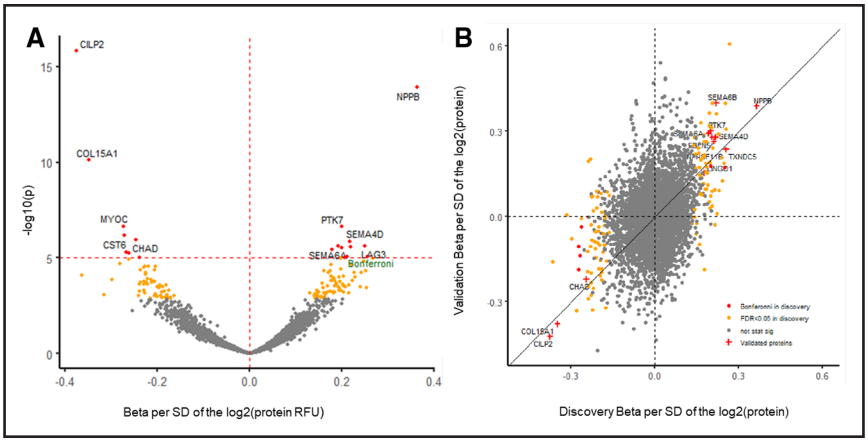

Figure 1. Volcano plot showing the estimates (beta coefficients) and significance (-log$_{10}$([*P* value]) of multivariable adjusted associations plasma proteins and heart failure among participants with diabetes (discovery sample; A) and scatter plot of beta coefficients for proteins associated with heart failure in the internal validation analysis among participants with diabetes (B), in the ARIC (Atherosclerosis Risk in Communties Study).

A, Red dots reflect statistical significance based on Bonferroni-corrected *P* value, orange dots reflect statistical significance based on false discovery rate (FDR) <0.05. **B**, Red plus signs reflect proteins identified in the discovery sample and confirmed in the validation sample. The estimates are adjusted for age, sex, race-center, estimated glomerular filtration rate, current smoking, body mass index, total cholesterol, high-density lipoprotein (HDL) cholesterol, systolic blood pressure, hypertension medication use, coronary heart disease, NT-proBNP (N-terminal pro-brain natriuretic peptide), and high-sensitivity troponin T (the latter 2 biomarkers were assessed using targeted assays). All Bonferroni significant associations on discovery were directionally concordant in replication.

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External Validation of Proteins Associated With Diabetes-Related Heart Failure

The 12 internally validated proteins associated with HF among individuals with diabetes were tested for external replication in the MESA study. The characteristics of the MESA participants are provided in Table S8. In MESA, there were 5233 individuals (mean age: 63±9.6 years, maximum follow-up: 19 years), among whom 663 had diabetes (85 HF cases), and 4570 had no diabetes (240 HF cases).

Of these 12 proteins available in MESA, among individuals with diabetes, all 12 proteins showed the same direction of association with incident heart failure, and 6 (50%) had FDR q<0.05 (Table S9). Among these proteins, NT-proBNP was also associated with heart failure among individuals without diabetes, while the other 5 were not, comprising 4 novel proteins (PTK7, CHAD, LINGO1, and FBLN5), and 1 protein (CILP2) previously known to be associated with HF among individuals with diabetes. Among MESA participants without diabetes, none of the proteins other than NT-pro-BNP (q=5×10^{-9}) was associated with HF incidence at FDR q<0.05. These 6 proteins did not show a statistically significant interaction by diabetes status for the HF outcome in the MESA population.

Pathways Analyses of the Proteins in Diabetes-Related HF

The IPA analysis, using a broader set of 134 proteins associated with diabetes-related HF (FDR q<0.05 in model 5), identified overrepresented biological pathways (Figure S1A; Table S10). The top 5 canonical pathways included the LXR/RXR (liver X receptor-retinoid X receptor) activation, acute phase response signaling, white adipose tissue browning, semaphorin neuronal repulsive signaling, and granulocyte adhesion and diapedesis pathways. The top upstream regulator for a network of proteins associated with incident HF among those with diabetes was TNF (tumor necrosis factor), a cytokine and regulator of inflammatory signaling, which was predicted to be highly activated in relation to HF-associated proteins in the setting of diabetes (Figure S1B; Table S11). Other top upstream regulators included activation of TRAF3IP2, an adapter protein CIKS involved in regulating responses to cytokines and plays a central role in inflammatory signals and stress, and inhibition of NEUROG1, a transcription regulator of numerous genes. Other upstream regulators included IL-17 A and APOE, which were also predicted to be highly activated in relation to HF-associated proteins among those with diabetes.

DISCUSSION

Principal Findings

In this study, using an aptamer-based proteomic profiling platform, we assessed the association of ≈5000 proteins with HF among individuals with diabetes in a large community-based sample of Black and White adults with over 20 years of follow-up. We identified 6 plasma proteins associated with incident HF among individuals with

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Table 2. Proteins Associated With Heart Failure Among Individuals With and Without Diabetes in the Discovery Sample–ARIC Visit 2–1990–1992

Symbol	Protein name	Diabetes		No diabetes	
		Hazard ratio (95% CI)	*P* value*	Hazard ratio (95% CI)	*P* value*
CILP2	Cartilage intermediate layer protein 2	0.69 (0.63–0.75)	1.6×10^{-16}	0.82 (0.76–0.87)	4.18×10^{-09}
NPPB	N-terminal pro-BNP	1.44 (1.31–1.58)	1.3×10^{-14}	1.35 (1.27–1.42)	1.17×10^{-24}
COL15A1	Collagen alpha-1(XV) chain	0.71 (0.64–0.78)	7.9×10^{-11}	0.80 (0.75–0.87)	8.97×10^{-09}
MYOC	Myocilin	0.76 (0.66–0.84)	2.2×10^{-07}	0.82 (0.77–0.88)	4.03×10^{-09}
PTK7	Inactive tyrosine-protein kinase 7	1.22 (1.13–1.32)	2.4×10^{-07}	0.98 (0.91–1.05)	5.07×10^{-01}
CST6	Cystatin-M	0.76 (0.68–0.85)	6.8×10^{-07}	0.90 (0.84–0.96)	2.01×10^{-03}
CHAD	Chondroadherin	0.78 (0.71–0.86)	1.1×10^{-06}	0.85 (0.80–0.90)	3.93×10^{-07}
SEMA4D	Semaphorin-4 d	1.24 (1.14–1.36)	1.4×10^{-06}	1.02 (0.95–1.09)	5.96×10^{-01}
LAG3	Lymphocyte activation gene 3 protein	1.28 (1.16–1.42)	2.3×10^{-06}	1.00 (0.94–1.06)	3.12×10^{-0}
SEMA6A	Semaphorin-6A	1.21 (1.12–1.31)	2.5×10^{-06}	0.97 (0.91–1.04)	4.53×10^{-01}
SEMA6B	Semaphorin-6B	1.24 (1.14–1.36)	2.6×10^{-06}	1.07 (1.00–1.14)	4.41×10^{-02}
COLEC11	Collectin-11	1.22 (1.12–1.33)	2.9×10^{-06}	0.98 (0.92–1.05)	6.34×10^{-01}
LINGO1	Leucine-rich repeat and immunoglobulin-like domain-containing Nogo receptor-interacting protein 1	1.19 (1.11–1.29)	3.7×10^{-06}	0.96 (0.90–1.02)	2.24×10^{-01}
MSMP	Prostate-associated microseminoprotein	0.76 (0.68–0.86)	4.8×10^{-06}	0.88 (0.82–0.93)	3.82×10^{-05}
NQO2	Ribosyldihydronicotinamide dehydrogenase [quinone]	0.77 (0.69–0.86)	5.9×10^{-06}	0.91 (0.86–0.97)	1.74×10^{-03}
TNFRSF11B	Tumor necrosis factor receptor superfamily member 11B	1.24 (1.13–1.36)	8.3×10^{-06}	1.09 (1.02–1.16)	1.06×10^{-02}
TXNDC5	Thioredoxin domain-containing protein 5	1.29 (1.15–1.44)	8.3×10^{-06}	1.04 (0.98–1.10)	1.89×10^{-01}
FAP	Prolyl endopeptidase FAP	0.79 (0.71–0.88)	9.1×10^{-06}	0.79 (0.74–0.84)	3.71×10^{-12}
FBLN5	Fibulin-5	1.23 (1.12–1.34)	9.1×10^{-06}	1.10 (1.04–1.16)	1.77×10^{-03}

Hazard ratios are adjusted for age, sex, race-center, estimated glomerular filtration rate, current smoking, body mass index, total cholesterol, HDL-cholesterol, systolic blood pressure, hypertension medication use, coronary heart disease, and NT-proBNP. The latter 2 biomarkers were assessed using targeted clinical immunoassays. All hazard ratios are expressed per SD of protein level. ARIC indicates Atherosclerosis Risk in Communities; HDL, high-density lipoprotein; and NT-proBNP, N-terminal pro-brain natriuretic peptide.

*The Bonferroni corrected significance level is $<10^{-5}$.

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diabetes after discovery, internal, and external validations. These associations were robust to a rigorous adjustment for confounders, Bonferroni, and FDR corrections. Of the 6 identified proteins, 4 were novel and specific to diabetes, and 1 had been previously associated with HF in the setting of diabetes (CILP2). The 5 proteins (CILP2, PTK7, CHAD, LINGO1, and FBLN5) other than NPBB were associated with HF only among those with diabetes and not among those without diabetes in the external validation. Three of the proteins (CILP2, PTK7, and CHAD) associated with diabetes-related HF were concordantly associated with lower levels in prevalent diabetes, higher risk of heart failure at higher levels, adding credence to the diabetes-specificity of the HF risk. Key biological pathways appeared to underpin diabetes-related HF, and these include alterations in lipid metabolism, adipose tissue browning, and systemic inflammation. Our findings provide important insights into the pathophysiological mechanisms leading to diabetes-related HF in humans, suggesting unique pathophysiological mechanisms (including alteration of lipid metabolism, inflammation, white adipose tissue browning, and semaphorin signaling) that could serve as a basis for developing novel prevention approaches for diabetes-related HF.

Comparisons With Other Proteomic Studies of Diabetes-Related Heart Failure

Our study complements and expands prior knowledge on the proteomic signatures of diabetes-related HF. We identified 5 proteins specifically associated with HF in the setting of diabetes, among which are 4 novel associations. A limited number of investigations have explored the specific proteomic signatures of diabetes-related HF, typically in cross-sectional studies.[9–13] A prior study (n=1572) of White adults using an immunoassay-based proteomic platform that measured 92 proteins (Olink) found that galectin-4, GDF (growth differentiation factor)-15, FABP (fatty acid binding protein)-4, and paraoxonase 3 were associated with prevalent diabetes-related HF.[9] Another small study of White participants (n=253) used SomaScan assays (4776 unique protein targets) and identified 3 proteins with significantly different expression in patients with diabetes and HF (vs no diabetes) with preserved ejection fraction, including alpha-1-microglobulin/bikunin precursor protein (higher among those with diabetes), as well as ApoM (apolipoprotein M) and CILP2 (both lower in diabetes).[10] A study of 1713 White individuals in whom 92 proteins were assessed (Olink platform)

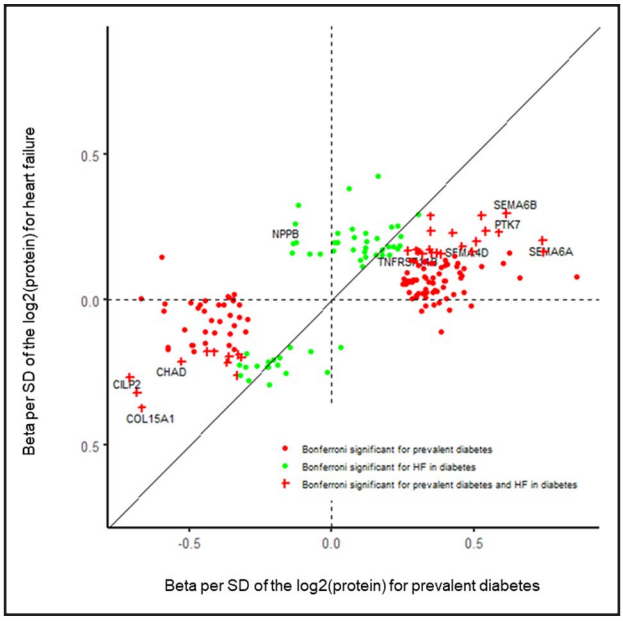

Figure 2. Scatter plot of beta coefficients for proteins associated with prevalent diabetes (in the discovery sample) vs proteins associated with incident heart failure (discovery sample) among individuals with diabetes in the ARIC (Atherosclerosis Risk in Communities Study).
Proteins found to be associated with incident heart failure among individuals with diabetes are shown.

showed that GAL4 (galectin-4) and CTSD (cathepsin D) were associated with diabetes-related HF.[11] Finally, in a study that assessed 92 proteins with an Olink platform among 5131 individuals with diabetes (70% Whites), HF hospitalization was associated with BNP (brain natriuretic peptide), Gal-9 (Galectin-9), VEGFD, FGF23, troponin, SCF (stem cell factor), and GIF (gastric intrinsic factor).[13] Although prior studies identified a few proteins similar to ours (CILP2 and NT-proBNP), these were generally small in size, quantified a lower number of proteins on a different proteomic platform, focused on White populations, identified proteins mostly related to inflammation or fibrosis, and did not always investigate the functional significance of the identified proteins through pathway analysis.[9–11,13] Furthermore, in some studies, biomarkers were measured after the development of both diabetes and HF, thus potentially clouding the temporal relationship of these conditions and introducing the possibility of reverse causation.[9,10] Indeed, processes linked to the development of HF may be distinct from pathophysiologic changes occurring after HF development. Some of the proteins identified in prior analyses did not show significant associations with HF in our investigation (eg, ApoM, GAL4, and CTSD),[9–13] possibly because of the greater extent of adjustments and more stringent threshold for significance in our approach.

Mechanisms

The proteins we identified represent key pathways, including: (1) alteration of lipid metabolism, (2) inflammation and apoptosis; (3) white adipose tissue browning, and (4) semaphorin signaling. Although the proteins in the semaphorin signaling pathway were validated internally but only nominally (P=0.04 for SEMA4D), the other identified pathways in our analyses are consistent with data from preclinical models of diabetes cardiomyopathy.[6,7] The alteration of lipid metabolism, as indicated by LXR/RXR activation, was a key pathway implicated

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Table 3. Association of the Validated Proteomic Markers of Diabetes-Related Heart Failure With Prevalent Diabetes–ARIC Visit 2–1990–1992

Gene	Protein name	Beta coefficient	SE	*P* value*	Significant association with prevalent diabetes
CILP2	Cartilage intermediate layer protein 2	−0.64	0.050	6.1×10^{-38}	Yes
NPPB	N-terminal pro-BNP	−0.05	0.049	2.7×10^{-01}	No
COL15A1	Collagen alpha-1 (XV) chain	−0.61	0.057	2.8×10^{-26}	Yes
PTK7	Inactive tyrosine-protein kinase 7	0.64	0.050	2.4×10^{-38}	Yes
CHAD	Chondroadherin	−0.49	0.049	9.2×10^{-24}	Yes
SEMA4D	Semaphorin-4 d	0.60	0.051	1.8×10^{-31}	Yes
SEMA6A	Semaphorin-6A	0.86	0.053	4.0×10^{-58}	Yes
SEMA6B	Semaphorin-6B	0.72	0.050	5.4×10^{-46}	Yes
LINGO1	Leucine-rich repeat and immunoglobulin-like domain-containing Nogo receptor-interacting protein 1	0.23	0.035	1.3×10^{-10}	Yes
TNFRSF11B	Tumor necrosis factor receptor superfamily member 11B	0.27	0.047	4.3×10^{-09}	Yes
TXNDC5	Thioredoxin domain-containing protein 5	0.10	0.043	2.6×10^{-02}	No
FBLN5	Fibulin-5	0.09	0.046	5.7×10^{-02}	No

Beta coefficients are adjusted for age, sex, race-center, estimated glomerular filtration rate, current smoking, body mass index, total cholesterol, HDL-cholesterol, systolic blood pressure, hypertension medication use, coronary heart disease, NT-proBNP, and high-sensitivity troponin T. The latter 2 biomarkers were assessed using targeted clinical immunoassays. All beta coefficients are expressed per SD of protein level. ARIC indicates Atherosclerosis Risk in Communities; HDL, high-density lipoprotein; and NT-pro-BNP, N-terminal pro-brain natriuretic peptide.

*The Bonferroni corrected significance level is $<10^{-5}$.

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in diabetes-related HF. Intracardiac LXR signaling has been implicated in the protection against myocyte hypertrophy and loss, fibrosis, and alteration in myocardial metabolism.[24] LXRs are also protective against diabetes through a modulation of glucose metabolism, β-cell insulin secretion, inflammatory signaling, and hyperglycemia-induced endothelial dysfunction.[24] In preclinical models, LXR agonists reduce plasma glucose, improve glucose tolerance and insulin sensitivity,[24] and attenuate cardiac dysfunction in type 2 diabetic db/db mice.[25]

The white adipose tissue browning pathway, to which NBPP belongs, was another key pathway described. Animal studies have shown that brown adipose tissue may limit myocardial injury.[26] Human studies showed that an increase in brown adipose tissue improves cardiac function via the release of lipokines,[27] as well as whole-body glucose homeostasis and insulin sensitivity.[28] Imaging studies in humans showed an inverse correlation between brown adipose tissue and diabetes on one hand, and with HF on the other hand.[29]

The upstream regulators identified as a top upstream regulator in our pathway analyses were mostly inflammatory cytokines, such as TNF, TRAF3IP2, and IL-17A. These findings are somewhat consistent with data from preclinical models implicating inflammation in the pathogenesis of diabetes-related cardiac remodeling. The data indicate a systemic inflammation in type 2 diabetes with increased circulating markers, such as TNF-α,[7] and intramyocardial induction of proinflammatory genes and proteins (eg, TNF-α and TGF-β) in models with type 2 diabetes.[7]

Clinical Implications

Novel protein biomarkers of diabetes-related HF can help to identify modifiable biological pathways and potential druggable targets relevant to HF. Our findings highlight an opportunity to improve strategies for preventing HF and identifying risk among individuals with diabetes. Indeed, higher levels of proteins positively associated with HF (more harmful associations) in diabetes and lower levels of proteins negatively associated with HF (less protective associations) can both serve as markers of the risk of disease worsening. Additional studies are warranted to assess whether the identified proteins may be targeted by lifestyle or pharmacological interventions for the prevention and management of HF in the setting of diabetes. SGLT2 (sodium-glucose cotransporter 2) inhibitors have been shown to reduce HF hospitalization,[30,31] through unclear mechanisms. Their impact on the identified diabetes-related HF pathways should be investigated. To that end, our study may be informative.

Strengths and Limitations

Our study has several strengths. First, our investigation was conducted in a large community-based sample including Black and White adults, with the quantification of >4500 proteins using a state-of-the-art platform, and extended follow-up for end points. Second, we conducted robust adjustment for covariates (including GFR estimated from both serum creatinine and cystatin C), relevant adjustments for the level of significance, internal and external replication, and an exploration of the causal links and mechanistic pathways. We adjusted for hs-cTnT and NT-pro-BNP from targeted immunoassays to account for subclinical myocardial damage and increased ventricular wall stress. We also accounted for CHD events before HF occurrence to isolate nonischemic pathways linked to HF development in diabetes.[32]

Our study has limitations. First, we did not have data on the HF subtypes, including HF with preserved ejection fraction or HF with reduced ejection fraction, possibly limiting our ability to detect proteomic association which are specific to both diabetes and HF subtypes. We focused on incident HF hospitalization, which does not include individuals with only an outpatient diagnosis of HF or excludes subclinical HF at baseline. We also lacked data on the various etiologies of HF, which would provide further mechanistic insight. Second, the power to detect interactions by diabetes status, especially in the external validation sample, which included fewer individuals overall, as well as fewer diabetes and heart failure cases, was limited. Because of the potentially limited power to detect interaction related to HF associations, we restricted our interaction tests to proteins shown to be significantly associated with HF among those with diabetes, which may be a limitation. We did not examine genetic effects since HF shows only weak genetic associations. Third, the MESA replication cohort did not include hemoglobin A1c, which may have limited our ability to detected difference between those with diabetes and without diabetes, including interaction tests. Fourth, we did not account for the severity of diabetes or the diabetes drugs in our analysis, which may have limited our ability to better understand the diabetes-HF proteomic link. Lastly, unmeasured factors (eg, changes in lifestyle factors and use of medications over time) might have affected the proteome and our outcome, and thus our results.

Conclusions

In summary, we identified and validated 6 circulating proteins associated with HF in diabetes; CHAD, FBLN5, LINGO1, and PTK7 were novel, while CILP2 was previously reported, and NT-proBNP is known to also be associated with HF among individuals without diabetes. Our findings highlight the importance of specific proteins and biochemical pathways, including TNF upregulation, accounting for excess diabetes-related HF risk.

ARTICLE INFORMATION

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Disclosures

Johns Hopkins University has signed a collaboration agreement with SomaLogic to conduct SomaScan of ARIC stored samples at no charge in exchange for the rights to analyze linked ARIC phenotype data. Dr Coresh is a scientific advisor to Soma Logic. Dr Walker is funded by the National Institute on Aging's Intramural Research Program. This study was funded, in part, by the National Institute on Aging's Intramural Research Program. The other authors report no conflicts.

Supplemental Material

Tables S1–S11
Figure S1

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